

weak entity set:

An entity set may not have sufficient attributes to form a primary key.

Such entity is called weak entity set.

As the weak entity do not have any primary key, they can not be identified on their own.

So they need to depend on other entity set called strong identity set.

Instead of having primary key, weak entity set has partial key. partial keys are set of attributes which help the weak entity to be identified.

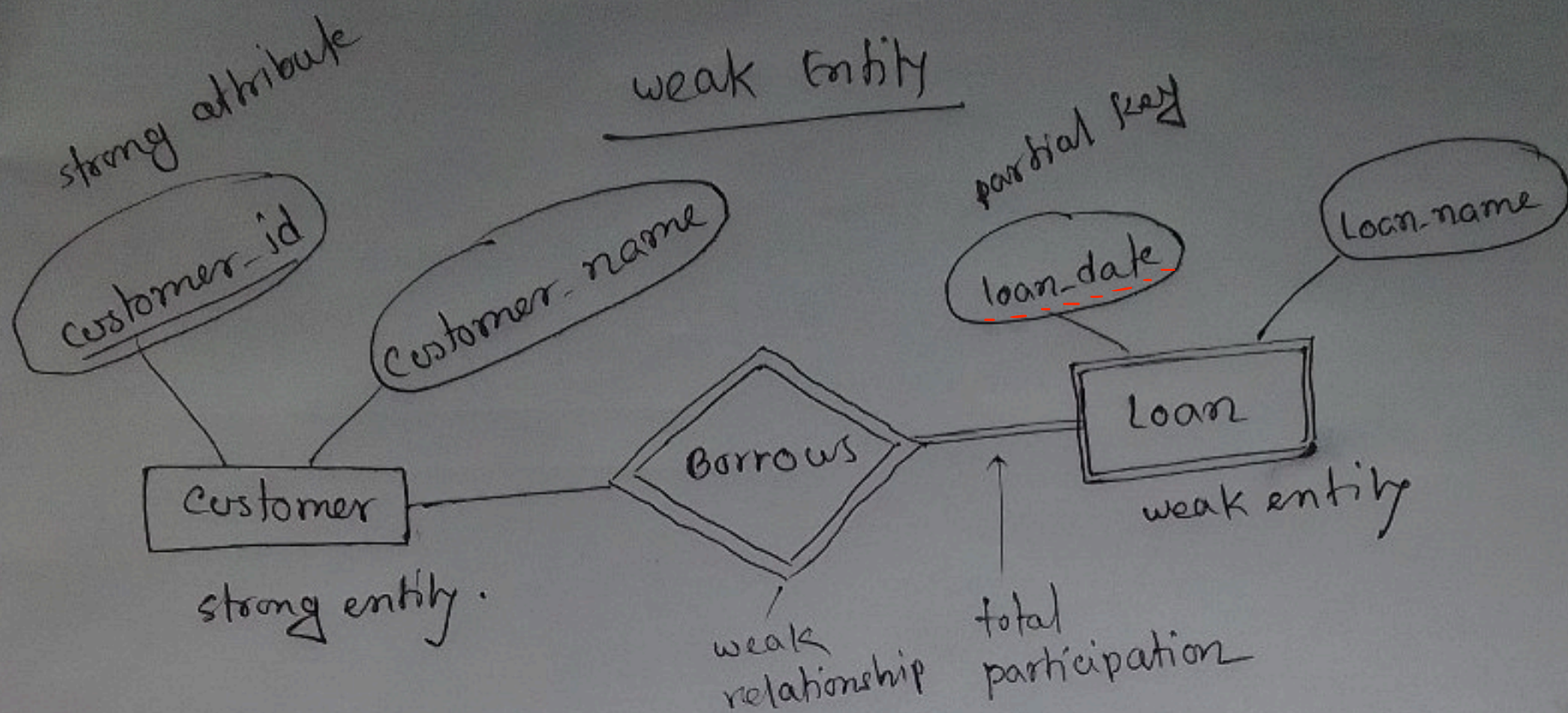
The relationship between weak and its corresponding strong entity is called an identifying relationship.

This identifying relationship must have total participation from the weak entity.

(i.e. every weak entity must be related to a strong entity)

weak entity sets are represented using a double rectangle in ER diagram.

The identifying relationship is represented using a double diamond.



Customer:

- strong entity
- primary key: customer-id

Customer is a strong entity because this entity has a unique identifier called customerID. This is called primary key.

Loan:

- weak entity
- partial key: loan-date

Loan-date, which cannot uniquely identify a Loan, because multiple customers can take Loan on the same day.

Therefore Loan-date is partial key, which can be matched with customerID to identify a Loan.